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Graded Assignment • 30 min

Due Mar 13, 11:59 PM CET

1. An event is: 1 point

- (●) any subset A of the sample space
- ( ) a probability
- ( ) a function which assigns probabilities to events

2. Probability... 1 point

- (●) must be non-negative
- ( ) must be negative
- ( ) can take any value in the interval $[-1,\ 1]$

3. Everyone will agree on the exact probabilities of various geopolitical events. 1 point

- ( ) True
- (●) False

4. When there are N equally likely outcomes in a sample space, where none of them agrees with some event A, then: 1 point

- ( ) $P(A) = 0$
- (●) $0 < P(A) < 1$
- ( ) $P(A) = 1$

5. The expectation of a (discrete) random variable X: 1 point

- ( ) is a probability-weighted average
- ( ) is a non-probability-weighted average
- (●) must be equal to a possible value of X

6. If we rolled a fair die a very large number of times, then we would expect the average score to be approximately: 1 point

- ( ) 2.5
- (●) 3.5
- ( ) 4.5

7. For the family of Bernoulli distributions, π is a: 1 point

- ( ) variable
- ( ) sample space
- (●) parameter

8. A Bernoulli random variable can take how many possible values? 1 point

- ( ) 0
- ( ) 1
- (●) 2

9. 1 point

If $X \sim \mathrm{Bin}(n,\ \pi)$ then $\mathrm{E}(X) = n\,\pi$.

- (●) True
- ( ) False

10. If $X \sim \mathrm{Bin}(n,\ \pi)$ then X is well-approximated by a $\mathrm{Poisson}(\lambda = n\,\pi)$ distribution when: 1 point

- ( ) n is small and π is small
- (●) n is large and π is small